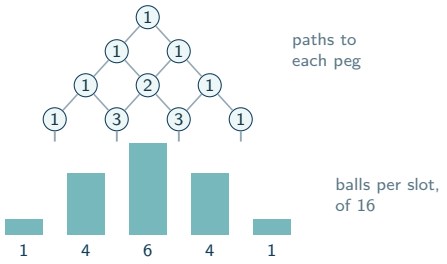
BINOMIAL ON ONE PAGE

From the Galton board to the binomial

Trees count paths, Pascal's triangle counts them for free, and the board runs the experiment.

1 / THE RULE AND THE PICTURE

Each peg sends a ball left or right, and the slot counts the rights



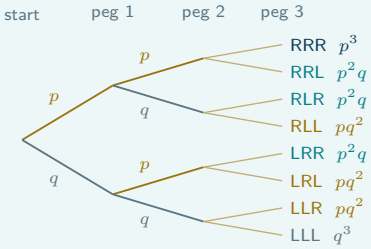
At each peg a ball goes right with probability p and left with $q = 1-p$, independently, so after n rows the slot is k, the number of rights. Routes to a slot are the ways to choose which k pegs deflect right, and Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ adds the two parents:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

For a fair board every route has probability 2^{-n}, so the slot probabilities are the Pascal row over 2^n.

2 / TREE DIAGRAM / THREE ROWS OF PEGS

Multiply along each path, then add the paths that reach the same slot

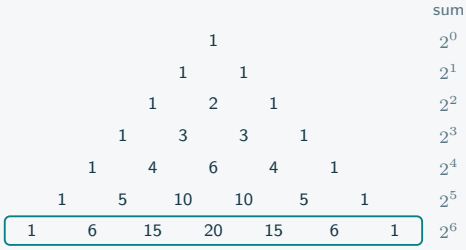


rights k	routes	$P(X=k)$	$p = 0.3$
3	1	p^3	0.027
2	3	$3p^2q$	0.189
1	3	$3pq^2$	0.441
0	1	q^3	0.343

Probabilities multiply along a path and routes to one slot add, which gives the counts $1, 3, 3, 1$. A fair board gives $\frac{1}{8}, \frac{3}{8}, \frac{3}{8}, \frac{1}{8}$, and tilting to $p = 0.3$ keeps the counts and shifts the weights.

3 / PASCAL'S TRIANGLE / PROBABILITY AND ODDS

The row gives the routes, 2^n gives the total, and the odds follow

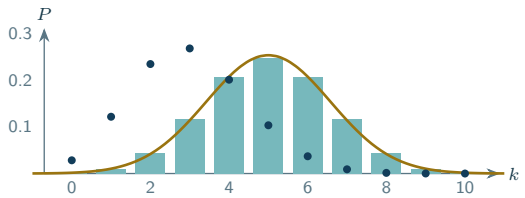


k	0	1	2	3	4	5	6
routes $\binom{6}{k}$	1	6	15	20	15	6	1
$P = \binom{6}{k}/64$	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$
odds for	1:63	3:29	15:49	5:11	15:49	3:29	1:63

Probability is favourable over total, $\binom{n}{k}/2^n$; the odds for are favourable to unfavourable, $\binom{n}{k} : (2^n - \binom{n}{k})$. The centre slot of a 6-row board has $P = \frac{5}{16}$ and odds $5 : 11$, and the two outer slots together have odds $1 : 31$.

4 / THE DISTRIBUTION / SHAPE, CENTRE AND SPREAD

With n pegs the slots follow $\mathrm{Bin}(n, p)$ with mean np and variance npq



Bars show the fair board with $n = 10$: mean $np = 5$ and $\sqrt{npq} \approx 1.58$, with the bell curve of the same mean and spread over them. The dots show $p = 0.3$, with mean 3.0 and $\sqrt{npq} \approx 1.45$, skewed. As n grows the spread rises as $\sqrt{n}$ and the bars settle onto the normal curve.

$$P(k) = \binom{n}{k} p^k q^{n-k}, \qquad \mu = np, \qquad \sigma^2 = npq$$

5 / SYNTHESIS

One count, three pictures: tree, triangle and board

picture	what it does	what it gives
tree diagram	lists every route, multiplying branch probabilities	the weight $p^k q^{n-k}$ of one route
Pascal triangle	counts the routes by adding parents	the coefficient $\binom{n}{k}$ without drawing the tree
Galton board	runs the experiment with many balls	the histogram that approaches $\mathrm{Bin}(n, p)$

ASIDE / WHY A BELL

Each peg adds an independent step, so variances add and the width grows as $\sqrt{n}$, the dice sheet's rule.

ASIDE / CHECK IN PYTHON

`[comb(6,k) for k in range(7)]` gives $1, 6, 15, 20, 15, 6, 1$, and `binom.pmf(1,3,0.3)` gives 0.441, the $3pq^2$ entry.

Convention. X counts rights, pegs are independent and $p + q = 1$; odds are favourable to unfavourable. Companions: the sqrt-time dice sheet.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames